

Mid Project Report

CS 896 – Data Science Capstone

**Project: Surface Water and Drinking
Water Contaminants**

Project Team – 10

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1. Project Objective:

The objective of this project is to analyze municipal drinking water contaminant data (CCR dataset) to understand patterns in contaminant concentrations, evaluate regulatory exceedances, and identify which contaminants pose higher risk to drinking water systems.

The analysis focuses on identifying trends over time, comparing contaminant behavior, and evaluating exceedance rates based on EPA Maximum Contaminant Level (MCL) standards. The original plan also included spatial analysis using watershed data; however, due to dataset limitations, the project is currently focused on statistical and contaminant-level analysis.

2. Dataset Overview

The dataset used in this project is the Municipal Drinking Water (CCR) dataset, which contains information about contaminant measurements reported by water systems. Key characteristics of the dataset include:

- Approximately 16,000 observations
- Multiple contaminants (e.g., Barium, Fluoride, Selenium, etc.)
- Concentration values reported in different units
- EPA Maximum Contaminant Level (MCL) values
- Reporting years for temporal analysis

The dataset includes multiple contaminants such as Barium, Fluoride, Selenium, Chromium, Nitrate, and others. Since the data was collected from different sources, it contained inconsistencies such as mixed units, text-based values, and inequality symbols.

3. Work Completed So Far

The project has progressed through multiple stages, starting from data cleaning to exploratory analysis and initial modeling.

3.1 Data Cleaning and Preprocessing

The initial phase of the project focused on preparing the dataset for analysis. This involved converting contaminant concentration values from text format into numeric format, as many values contained symbols such as inequality signs and inconsistent formatting.

Measurement units such as ppm, mg/L, µg/L, and ppb were standardized to a common unit to ensure consistency across observations. This step was essential for enabling meaningful comparisons between contaminant concentrations.

Additionally, a regulatory exceedance indicator variable was created to identify whether contaminant concentrations exceeded EPA Maximum Contaminant Level thresholds. Outlier detection was also performed to identify extreme values, which were retained for analysis but flagged for further review.

3.2 Exploratory Data Analysis (EDA)

Following data preparation, exploratory data analysis was conducted to understand the structure and patterns within the dataset.

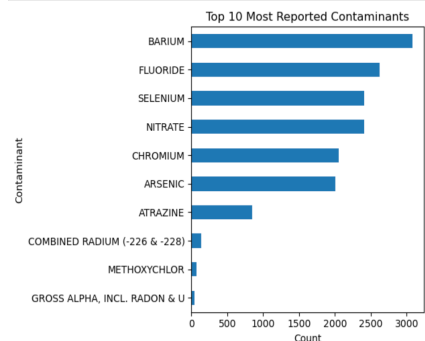
The analysis included:

- Identifying the most frequently reported contaminants
- Evaluating exceedance rates across contaminants
- Examining trends in exceedance rates over time
- Analyzing the distribution of contaminant concentrations

Top 10 Most Reported Contaminants (Bar chart)

```
top10 = df["Contaminant"].value_counts().head(10)

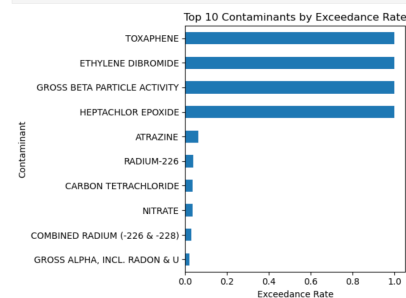
plt.figure()
top10.sort_values().plot(kind="barh")
plt.title("Top 10 Most Reported Contaminants")
plt.xlabel("Count")
plt.ylabel("Contaminant")
plt.tight_layout()
plt.show()
```



Top 10 by Exceedance Rate (Bar chart)

```
ex_rate = (
    df.groupby("Contaminant")["Exceeds_MCL"]
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

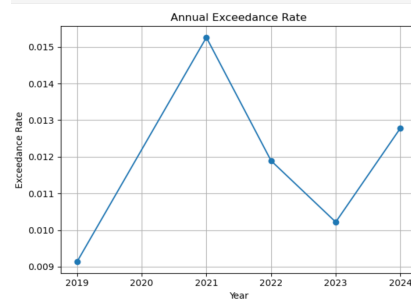
plt.figure()
ex_rate.sort_values().plot(kind="barh")
plt.title("Top 10 Contaminants by Exceedance Rate")
plt.xlabel("Exceedance Rate")
plt.ylabel("Contaminant")
plt.tight_layout()
plt.show()
```



Annual Exceedance Trend (Line graph)

```
annual = (
    df.dropna(subset=["Year_num"])
    .groupby("Year_num")["Exceeds_MCL"]
    .mean()
    .sort_index()
)

plt.figure()
annual.plot(marker="o")
plt.title("Annual Exceedance Rate")
plt.xlabel("Year")
plt.ylabel("Exceedance Rate")
plt.grid(True)
plt.tight_layout()
plt.show()
```



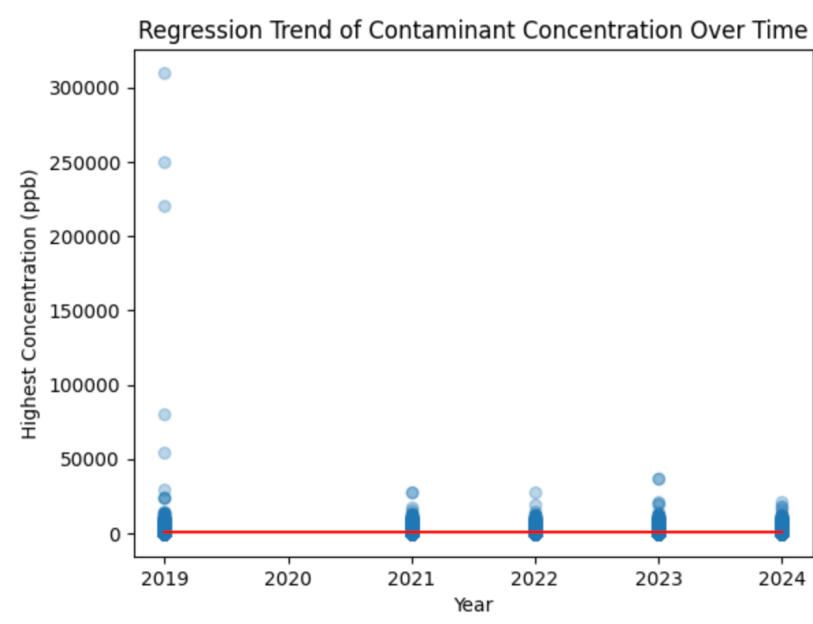
The analysis showed that most contaminant values are concentrated at lower levels, while a small number of observations have very high concentrations. This resulted in a right-skewed distribution, indicating high variability across contaminants.

These analyses provided a foundational understanding of contaminant behavior and highlighted key variables for further investigation.

3.3 Preliminary Statistical Modeling

To evaluate whether contaminant concentrations change over time, a linear regression model was implemented using reporting year as the predictor variable and contaminant concentration as the response variable.

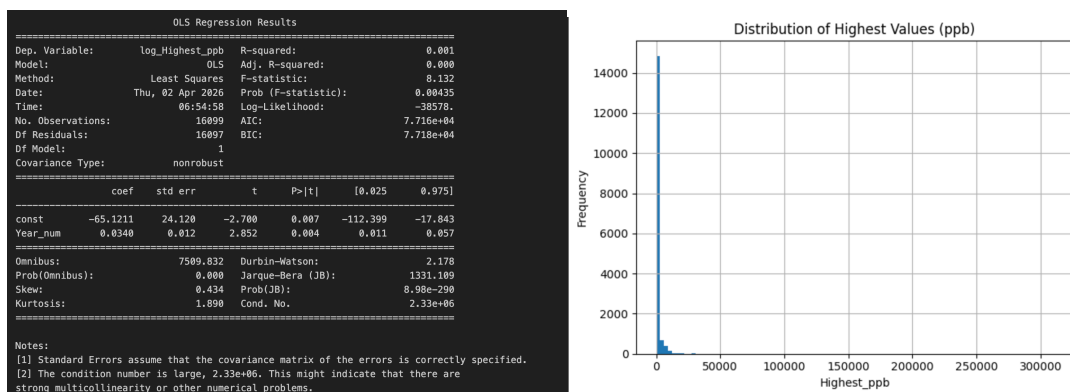
The results indicated that reporting year does not significantly explain variation in contaminant concentrations when all contaminants are analyzed together ($R^2 \approx 0$). This outcome is expected due to the presence of multiple contaminant types with varying concentration ranges. A log transformation of concentration values was also applied to reduce skewness and improve model stability



These results suggest that temporal trends may be more meaningful when analyzed separately for individual contaminants.

3.4 Log Transformation

To address skewness in the data, a log transformation was applied to the concentration values.



The log-transformed regression showed a slight positive trend with a statistically significant p-value. However, the R^2 value remained very low, indicating that time is still not a strong predictor of contaminant concentration.

This suggests that variability is primarily driven by differences between contaminants rather than time.

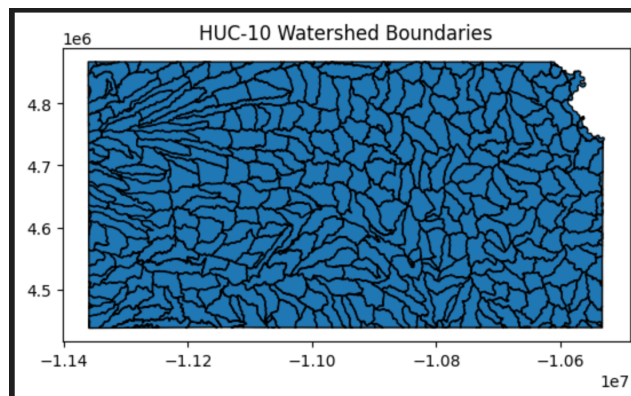
The scatter plot shows that most contaminant concentration values are heavily clustered near the lower range (close to 0 ppb), while a small number of observations in 2019 have extremely high values (above 200,000 ppb). These extreme values create a strong right-skew in the data.

The regression line appears almost flat across the years, indicating that there is no clear increasing or decreasing trend in contaminant concentration over time when using the raw values. This suggests that variation in concentration is dominated by extreme values and differences between contaminants rather than changes over time.

3.5 Spatial Data Integration and Contaminant Analysis

In the most recent phase, spatial analysis was introduced by incorporating watershed boundary data (HUC-10 regions) using GeoPandas.

We believe the watershed dataset provides geographic boundaries that allow for spatial analysis of contaminant behavior across different hydrologic regions.

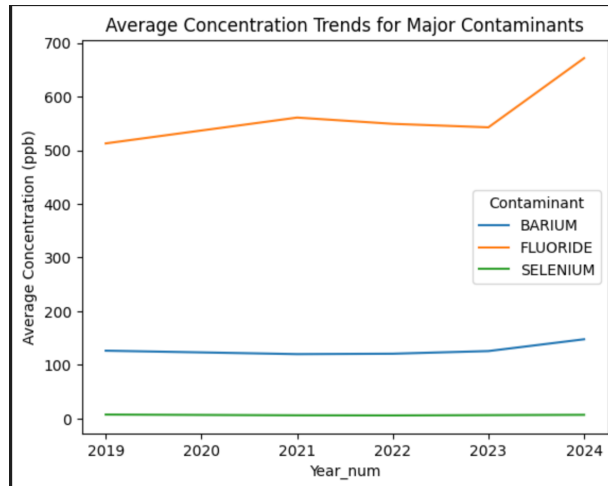


However, the contaminant dataset does not contain latitude or longitude information, making it impossible to perform spatial joins between contaminant data and watershed regions.

As a result, spatial analysis could not be completed.

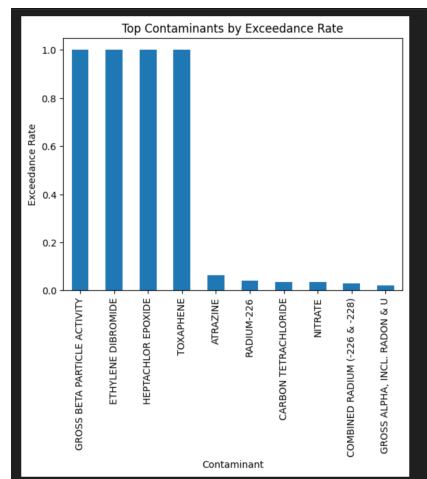
Contaminant-specific trend analysis was also conducted, revealing that:

- Fluoride shows relatively higher concentration levels and recent increases
- Barium remains relatively stable over time
- Selenium shows low and stable concentrations

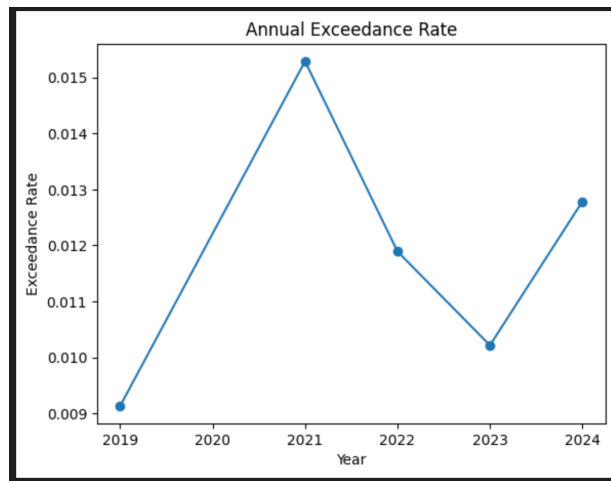


The contaminant-specific trend plot shows how average concentration levels change across reporting years for the most frequently reported contaminants in the dataset. Fluoride exhibits the highest average concentration values and shows a noticeable increase in the most recent year. Barium concentrations remain relatively stable across the reporting period with only minor fluctuations. Selenium concentrations remain comparatively low and show little variation over time. Analyzing contaminants separately helps reveal patterns that may be hidden when all contaminants are analyzed together in a single regression model.

Exceedance Rate by Contaminant



The exceedance rate analysis identifies contaminants that most frequently exceed EPA Maximum Contaminant Level thresholds. The above contaminants exhibit very high exceedance rates, indicating potential regulatory concern, while others show substantially lower exceedance frequencies. This analysis helps distinguish contaminants that pose higher compliance risks and provides guidance for prioritizing contaminants for further investigation.



Exceedance rate analysis identified contaminants that frequently exceed regulatory limits, highlighting potential areas of concern. Additionally, annual exceedance trends were analyzed to understand how regulatory violations change over time.

4. Work Completed This Week

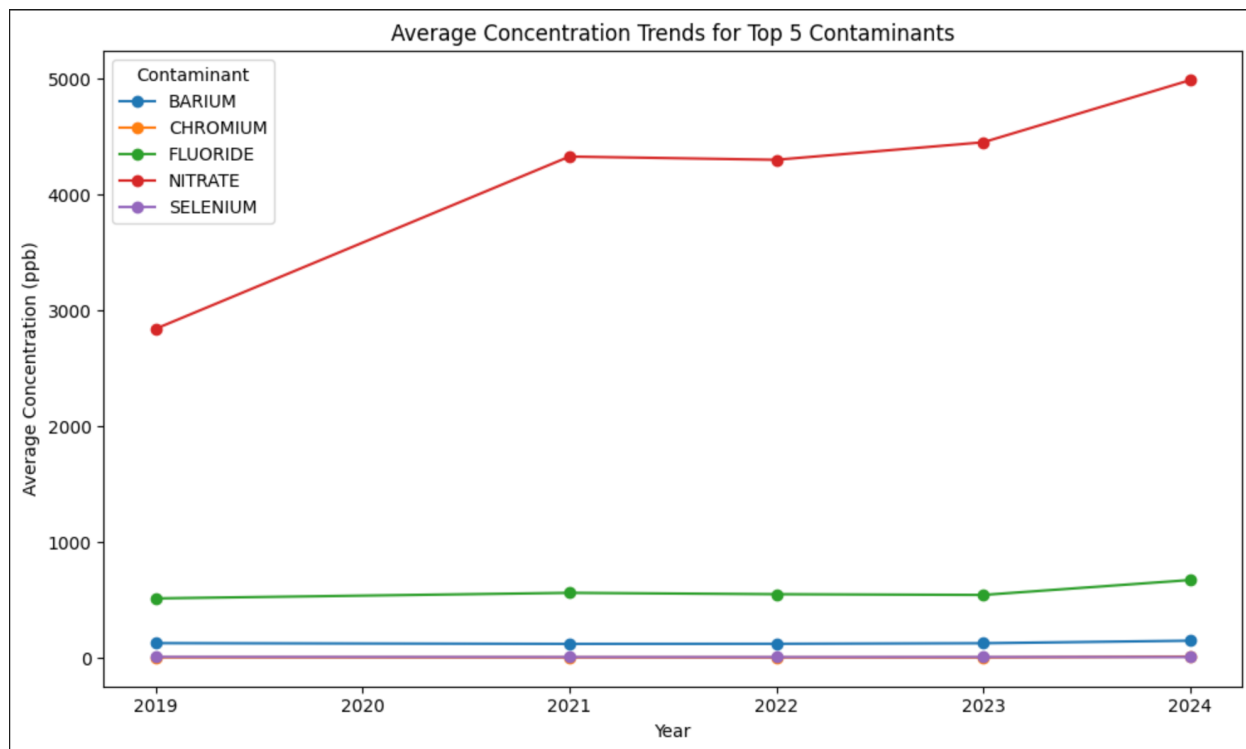
This week focused on deeper analysis at the contaminant level rather than treating all contaminants as a single group.

4.1 Contaminant Trend Analysis

The top 5 most frequent contaminants were selected, and their average concentration trends over time were analyzed.

The results show that Nitrate has significantly higher concentration values compared to other contaminants and shows an increasing trend over time. Fluoride shows moderate concentration levels with slight variation, while Barium and Chromium remain relatively stable. Selenium has consistently low concentration values.

This analysis highlights that each contaminant behaves differently, and combining all contaminants in a single model hides these patterns.



The plot shows clear differences in concentration levels across contaminants. Nitrate has significantly higher concentration values compared to all other contaminants, increasing from around 2800 ppb in 2019 to nearly 5000 ppb in 2024. This indicates a noticeable upward trend over time.

Fluoride shows moderate concentration levels and remains relatively stable across years, with a slight increase in the most recent year. Barium and Chromium have lower concentration values and show only minor variation over time.

Selenium remains at very low concentration levels throughout all years with almost no visible change. Overall, the plot highlights that Nitrate behaves very differently from the other contaminants and dominates the overall concentration trend.

4.2 Outlier Rate Analysis

Outlier rates were calculated for each contaminant to understand variability.

Radium-228 showed the highest outlier rate; however, it has a small number of observations, which limits reliability. Atrazine and Barium show moderate outlier rates with larger observation counts, making them more meaningful. Fluoride shows a relatively low outlier rate, indicating stable behavior.

The analysis shows that outlier interpretation should consider both rate and sample size.

	Outlier_Count	Observation_Count	Outlier_Rate
Contaminant			
RADIUM-228	5	28	0.178571
CARBON TETRACHLORIDE	2	28	0.071429
ATRAZINE	54	846	0.063830
BARIUM	182	3086	0.058976
SELENIUM	130	2408	0.053987
CHROMIUM	106	2057	0.051531
FLUORIDE	115	2627	0.043776
COMBINED RADIUM (-226 & -228)	6	141	0.042553
GROSS ALPHA, INCL. RADON & U	2	48	0.041667
ARSENIC	67	2008	0.033367

The table shows that Radium-228 has the highest outlier rate (approximately 17.9%), even though it has a relatively small number of observations (28). This suggests that a larger proportion of its values are extreme compared to other contaminants.

Carbon Tetrachloride and Atrazine also show relatively higher outlier rates, but Atrazine has a much larger number of observations (846), making its outlier behavior more reliable.

Barium, Selenium, and Chromium have moderate outlier rates with large observation counts, indicating consistent but controlled variability. Fluoride has a lower outlier rate despite having a large number of observations, suggesting more stable concentration levels.

Overall, the results show that some contaminants exhibit extreme values more frequently, but interpretation must consider the number of observations to avoid misleading conclusions.

4.3 Final Summary Analysis

A final summary table was created combining:

- average concentration
- exceedance rate
- outlier count
- observation count

The results show that some contaminants have very high exceedance rates but are based on very few observations. In contrast, contaminants like Atrazine have moderate exceedance rates with larger datasets, making them more important from a risk perspective.

Nitrate shows very high concentration levels but relatively lower exceedance rates, indicating that high concentration alone does not necessarily imply regulatory violation.

Contaminant	Average_Concentration_ppb	Exceedance_Rate	Outlier_Count	Observation_Count	Outlier_Rate
HEPTACHLOR EPOXIDE	210.000000	1.000000	0	1	0.000000
ETHYLENE DIBROMIDE	68.000000	1.000000	0	1	0.000000
TOXAPHENE	21.000000	1.000000	0	1	0.000000
GROSS BETA PARTICLE ACTIVITY	6.600000	1.000000	0	4	0.000000
ATRAZINE	0.972453	0.062648	54	846	0.063830
RADIUM-226	1.237037	0.041667	0	27	0.000000
CARBON TETRACHLORIDE	1.435000	0.035714	2	28	0.071429
NITRATE	3912.938487	0.034497	22	2406	0.009144
COMBINED RADIUM (-226 & -228)	1.491986	0.028369	6	141	0.042553
GROSS ALPHA, INCL. RADON & U	4.570833	0.021739	2	48	0.041667

The summary table highlights important differences between contaminants in terms of concentration, exceedance rate, and variability.

Some contaminants such as Heptachlor Epoxide, Ethylene Dibromide, Toxaphene, and Gross Beta Particle Activity show an exceedance rate of 1.0. However, each of these has only one or a few observations, which makes these results less reliable for drawing strong conclusions.

Atrazine has a moderate exceedance rate (~0.063) with a large number of observations (846), making it a more meaningful contaminant to monitor. Nitrate stands out with an extremely high average concentration (~3912 ppb) but a relatively low exceedance rate (~0.034), indicating that high concentration does not always imply frequent regulatory exceedance.

Carbon Tetrachloride and Combined Radium show moderate exceedance rates with smaller observation counts, while Barium and Selenium (not shown in top rows) typically have larger datasets with more stable behavior.

Overall, this table shows that both concentration levels and exceedance rates must be considered together, and results based on very small sample sizes should be interpreted with caution.

5. Limitations

One of the major limitations of this project is the absence of geographic information such as latitude and longitude in the dataset.

Due to this limitation, spatial analysis and linkage between surface water sources and drinking water systems could not be performed. This restricts the ability to identify geographic contamination patterns and hotspots.

6. Final Evaluation Goals – Progress and Findings

Goal 1: Analyze contaminant impact on drinking water Status: Completed

Finding: Contaminant behavior varies significantly, and time alone does not explain concentration changes.

Goal 2: Identify high-risk contaminants Status: Completed

Finding: Risk depends on a combination of concentration, exceedance rate, and variability rather than a single factor.

Goal 3: Link surface water to drinking water Status: Not Completed

Reason: Lack of geographic data prevented spatial linkage.

7. Work Remaining

Although the main analysis has been completed, several important steps remain.

The next step is to refine the interpretation of results and clearly identify the most important contaminants based on combined metrics such as concentration, exceedance rate, and variability.

The findings need to be organized in a clear and structured manner for final presentation. This includes selecting the most relevant graphs and explaining insights in a concise and understandable way.

Additionally, the limitations of the study, especially the inability to perform spatial analysis, need to be clearly explained as part of the final discussion.

Overall, the remaining work focuses on strengthening the interpretation and presenting the results effectively.

8. Conclusion

The project successfully cleaned and analyzed the CCR dataset to understand contaminant behavior in drinking water systems.

The results show that contaminant patterns vary widely across types, and combining all contaminants can hide important insights. Time alone does not explain variation, and contaminant-specific analysis is necessary to understand risk.

Although spatial analysis could not be performed due to data limitations, the project provides meaningful insights into contaminant trends, exceedance behavior, and variability.

The final stage of the project will focus on consolidating findings and presenting them clearly.

Overall, the results indicate that contaminant-specific analysis is more meaningful than aggregate analysis, as different contaminants exhibit very different behavior in terms of concentration, variability, and regulatory exceedance.